

1) Make sure Cloudflare can actually proxy **private.example.com**

- In Cloudflare DNS, create **private** record (usually CNAME or A) and set it to **Proxied (orange cloud)**.
- Your origin can be:
 - Public (regular web server behind Cloudflare), or
 - Private (behind a firewall) via **Cloudflare Tunnel**.

If you don't have an origin reachable, Cloudflare can't serve the hostname. Access is an auth gate, not hosting by itself.

2) Create an Access Application for the hostname

In Cloudflare Zero Trust:

- **Access** → **Applications** → **Add an application**
- Type: **Self-hosted**
- Domain: **private.example.com** (or **private.example.com/***)

3) Add Okta as the IdP

In Cloudflare Zero Trust:

- **Settings** → **Authentication** → **Login methods** → **Add**
- Choose **Okta**
- Cloudflare will give you values you'll paste into Okta (Redirect URI / Callback URL etc.)

In Okta (Admin):

- Create a new **OIDC app integration** (OpenID Connect)
 - Type: **Web**

- Set **Sign-in redirect URIs** to what Cloudflare provides
- Assign users/groups to the Okta app
- Copy:
 - **Okta domain** (your org URL)
 - **Client ID**
 - **Client Secret**
- Paste those into Cloudflare's Okta login method setup

4) Write your Access policy (who is allowed)

In the Access Application:

- **Policies** → **Add a policy**
- Example:
 - **Allow** if user is in Okta group **PrivateAppUsers**
 - (Optional) Require MFA (handled by Okta sign-on policy)
 - (Optional) require device posture, IP ranges, country, etc.

5) What users see

User hits **https://private.example.com** →

- Cloudflare redirects to Okta →
- user signs in (SSO/MFA) →
- Cloudflare sets an Access session →
- request is allowed through to your app.

✓ Your application doesn't need to "know" about Okta at all in this model — Cloudflare enforces the login at the edge.

Option B: Your app uses Okta directly (Cloudflare just proxies)

Use this if you need the **app itself** to receive tokens/claims and do role-based auth internally.

- You still create an Okta OIDC/SAML app integration
- Your